UI Design Specification

UI Design Specification | Version 1.0 | 19 October 2026

1. Direction

The first version read as machine-made for four reasons: a heavy black box dominated the first screen, everything was a flat color block, the result number shouted like a headline, and the copy described the implementation instead of helping the person. The fix is warmth and restraint: lighter surfaces, thin borders, a few physical details, one confident number, and copy written for a person.

2. Color tokens

Contrast ratios are approximate, computed with the WCAG relative-luminance formula. Verify with your own checker before shipping.

Token	Hex	Use	Contrast
paper	#F4F1EA	Page background	n/a
surface	#FBF8F2	Cards	n/a
ink	#231F1B	Primary text	about 14.5:1 on paper
ink-2	#5C544B	Secondary text	about 6.6:1 on paper, 7:1 on surface
accent	#B4451C	Primary button, check marks, progress	about 4.9:1 on paper, 5.2:1 on surface
on-accent	#FFFFFF	Text on accent	about 5.5:1
accent-press	#983A17	Pressed and hover state of accent	higher than accent
line	rgba(35,31,27,0.10)	Hairline borders (decorative)	n/a
line-strong	rgba(35,31,27,0.18)	Unselected radio outline	n/a

Do not use lighter greys for text. #7A7167 on paper is about 4.25:1 and fails AA for normal text.

3. Typography

- **Display:** Fraunces (variable, optical-size axis), weight 400-500, tracking -0.02em. Warm and literary without reading as a newspaper.
- **UI and body:** Inter, weights 400-600, tracking -0.005em on buttons.
- **Result number:** Fraunces 400, tabular figures, with the unit set small in Inter beside it. Large because it is the answer, light so it reads as elegant.
- Labels are sentence case. No all-caps labels.

Role	Font	Size / line height	Weight
Home headline	Fraunces	40 / 42	500
Result number	Fraunces	88 / 88	400
Wordmark	Fraunces	18 / 24	500
Body	Inter	15 / 22	400
Secondary body	Inter	13-14 / 18-21	400

Role	Font	Size / line height	Weight
Label	Inter	13 / 18	500

4. Spacing, radii, elevation, materials

Item	Value
Spacing grid	4 pt. Screen padding 20. Related items 12. Sections 24-32.
Radii	Viewfinder 28. Card and mode card 24. Button 16. Stat tile 16. Progress track full.
Card elevation	1px hairline border + 1px white inner top highlight + ambient shadow (0 8px 24px -12px rgba(35,31,27,0.18)).
Recessed areas	Inner shadow (inset 0 2px 6px rgba(35,31,27,0.07)). Used by the viewfinder, progress track and stat tiles.
Background texture	SVG fractal noise at about 5% opacity, multiply blend, fixed. Felt, not seen.
Screen width	Optimized for 390 x 844. Content column max 430 wide, centered.

5. Components

Component	Spec	States
Viewfinder	Cream inset surface, aspect 3:4, four corner brackets, faint dashed body silhouette in the idle state. Instructions live below it as a checklist, not inside it. When a photo is present the surface becomes the image and the brackets stay.	idle, photo present
Primary button	Accent fill, 48 high, 16 radius, 1px inner top highlight, short colored shadow, white label.	default, hover/pressed (accent-press), disabled (40% opacity)
Secondary button	Surface fill, hairline border, ink label.	default, hover, pressed
Ghost button	No fill, ink-2 label.	default, hover
Mode card	Radio semantics. Icon, title, one-line note, check circle. Selected state uses a 2px accent ring plus a filled check, never color alone.	unselected, selected, pressed (scale 0.98)
Progress bar	8 high, inset track, accent fill, spring-animated width.	idle, animating, complete
Processing steps	Three rows: circle becomes accent check on completion.	pending, active, done
Stat tile	Label, tabular value, inset, 16 radius.	static
Result card	One surface: label, number + unit, imperial + range, quality bar and caveat, two stat tiles, one-line note.	counting, settled

6. Screens

6.1 Home

- Top bar: ruler icon + wordmark.
- Headline "Your height, from one photo." and one supporting sentence (max 32 characters per line).

- Radio group of three mode cards: Proportions, Horizon, Reference. Default: Proportions.
- Bottom: full-width Continue button, then the privacy line.

6.2 Capture

- Top bar: back button (hidden while processing) and the selected mode name.
- Viewfinder, then a three-item checklist.
- Two secondary buttons side by side: Take photo, Choose photo. Ghost button: Use a sample.
- Primary Measure my height, disabled until a photo exists. Privacy line below.
- On press, the checklist and buttons cross-fade into the Processing view: progress bar and three steps, announced politely to screen readers.

6.3 Result

- Card enters with a 220 ms fade and 12 px rise.
- Number counts from 0 to the result. Imperial value and range follow in the same line.
- Quality row: "Clean shot" on the left, "Photo quality 92/100" on the right, then the bar and the caveat.
- Two stat tiles: Head to toe, Head height. One line about heads tall.
- Bottom: Measure again (primary) and the privacy line.

7. Microcopy

Original	Final copy
Proportion estimate	(removed; headline does the job)
Estimate standing height from one photo. No reference object, only how many head lengths fit in the body.	Stand tall, take one photo, and get an estimate in seconds. No tape needed.
Head to feet, in frame	Head to toe in the frame
Stand straight, face the camera, and leave a little room above your hair and below your shoes.	Face the camera and stand tall. Leave a hand's width above and below.
Upload / Take photo / Try a sample photo / Mark points yourself	Choose photo / Take photo / Use a sample / Adjust points (P1)
Estimate height	Measure my height
Pose detection runs in the browser. The photo is not uploaded.	Processed on your device. Your photo never leaves your phone.
Estimated height	You're about
Good pose, Pose quality 92 / 100	Clean shot, Photo quality 92/100
This scores the pose... not an accuracy percentage...	This rates the photo, not the answer. Even a clean photo can be off by several centimetres.
Measured stature is 7.3 heads. The canonical constant is 7.5.	You measure 7.3 heads tall. Classical art uses 7.5.
(new) range	Likely 161-173 cm
(new) error	Your feet are cut off. Step back or tilt the phone down.

Copy rules: sentence case, plain verbs, active voice. A button names what happens. Errors say what went wrong and how to fix it, and never apologize.

8. Interaction and motion

Moment	Behavior
Button press	Scale to 0.97 with a spring (stiffness 500, damping 30) and darken to accent-press. Springs back on release. Android: light haptic tick.
Mode card press	Scale to 0.98, same spring.
Progress bar	Width animates with a spring (stiffness 140, damping 22), a slight overshoot that settles. Reduced motion: snap to value.
Processing steps	Each step's circle fills with an accent check on completion. Tie to real pipeline stages once they exist.
Result count-up	0 to value over 1.2 s, ease-out cubic-bezier(0.16, 1, 0.3, 1), tabular figures so width does not jitter. Reduced motion: show the final value. Screen readers get the final value as static text.
Screen transition	220 ms fade with a 12 px vertical shift.

Motion answers a person's action or marks the result arriving. There are no decorative looping animations.

9. Accessibility

- All text meets WCAG AA (4.5:1 normal text, 3:1 large text) with the tokens above.
- Selection and status never rely on color alone (icon, text, ring thickness).
- Interactive targets at least 44 x 44 px.
- Visible focus ring: 2px accent with a 3px offset.
- Respect the reduced-motion setting everywhere.
- The result is exposed as one text value, for example "Estimated height 167 centimetres, likely between 161 and 173."
- Camera guidance is available as text, not only as overlay graphics.
- Progress bars expose role, min, max and current value.

10. Mapping to Jetpack Compose

- Tokens become a custom ColorScheme and Typography. Avoid default Material colors and shapes.
- A framer-motion spring (stiffness k, damping c) maps to spring(dampingRatio = c / (2 * sqrt(k)), stiffness = k). Press: ratio about 0.67 at stiffness 500. Progress: about 0.93 at stiffness 140.
- Count-up: Animatable or animateIntAsState with tween(1200, easing = CubicBezierEasing(0.16f, 1f, 0.3f, 1f)).
- Card layers: Modifier.shadow, border, and a drawBehind highlight line.
- Fonts: bundle Fraunces and Inter as app resources, because Google Fonts CSS does not apply in Compose.

11. Known gaps in the web prototype

MeasureMe.jsx is a visual and interaction reference, not production code. When a coding agent tweaks it, these are the open items:

1. Result values (167 cm, plus or minus 6, quality 92, 7.3 heads, 1,400 px, 193 px) are hard-coded placeholders. The plus or minus 6 is an invented stand-in, not a real error bound.
2. Processing is a timer, not the real pipeline.
3. No live camera view. Take photo opens the system file picker with a camera hint.
4. Missing: Adjust points, unit toggle (cm and ft/in), history and growth chart, camera-height input for Horizon, A4 placement guide for Reference, error states for the failure reasons in the Technical Design Document.
5. Fraunces and Inter are loaded from Google Fonts. The Android app must bundle them.